

HELSINKI VANTAA, Finland

WMO: 029740

Lat: 60.327N

Lon: 24.957E

Elev: 54

StdP: 100.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.5	-18.2	-24.2	0.4	-21.1	-20.9	0.6	-18.0	11.4	-0.3	10.5	-0.5	2.9	10	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	27.2	18.4	25.6	17.4	23.9	16.6	19.9	24.9	18.7	23.5	17.7	22.1	4.6	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	21.8	17.0	12.2	20.6	16.0	11.4	19.5	57.1	24.9	53.3	23.5	50.2	22.1	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.3	8.3	DB	-24.5	29.4	4.2	1.8	-27.6	30.7	-30.1	31.8	-32.5	32.8	-35.5	34.1	(4)
(5)			WB	-24.8	21.4	4.2	1.0	-27.8	22.1	-30.2	22.6	-32.6	23.2	-35.6	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-4.9	-5.3	-1.8	4.5	10.4	14.9	18.1	16.5	11.7	5.8	1.2	-2.2	(6)	
	DBStd	9.18	6.02	6.18	4.44	3.81	4.07	3.24	2.95	2.85	3.26	3.90	4.41	5.50	(7)	
	HDD10.0	2275	462	429	366	170	44	2	0	0	21	138	264	379	(8)	
	HDD18.3	4637	720	663	624	416	248	113	42	71	200	389	514	638	(9)	
	CDD10.0	738	0	0	0	5	56	148	250	203	71	7	0	0	(10)	
	CDD18.3	59	0	0	0	0	2	9	33	14	1	0	0	0	(11)	
	CDH23.3	470	0	0	0	0	22	76	271	101	1	0	0	0	(12)	
	CDH26.7	73	0	0	0	0	2	7	53	11	0	0	0	0	(13)	
(14) Wind	WSAvg	4.1	4.3	4.2	4.2	3.9	4.0	4.0	3.8	3.6	3.9	4.2	4.4	4.6	(14)	
(15) Precipitation	PrecAvg	662	47	33	33	36	36	52	68	79	68	76	72	59	(15)	
	PrecMax	903	105	112	78	101	86	124	201	192	150	193	216	168	(16)	
	PrecMin	422	8	3	3	3	3	12	4	9	12	16	5	8	(17)	
	PrecStd	120	25	22	16	21	19	29	43	39	35	42	39	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	5.5	11.1	19.6	26.1	27.7	30.2	28.2	22.2	15.1	10.1	8.1	(19)	
		MCWB	4.5	3.3	6.5	11.0	16.0	17.6	19.6	18.9	15.7	11.9	9.2	7.4	(20)	
	2%	DB	3.9	3.3	7.8	16.4	23.0	25.5	28.1	26.0	20.0	13.4	9.0	6.7	(21)	
		MCWB	3.2	2.4	3.9	9.1	14.2	16.6	19.4	18.0	15.1	11.5	8.3	6.0	(22)	
	5%	DB	2.8	2.5	5.2	13.7	20.6	23.6	26.3	24.1	18.2	12.2	8.0	5.4	(23)	
		MCWB	2.4	1.8	2.5	7.9	12.8	15.9	18.2	16.9	14.2	10.9	7.4	4.8	(24)	
	10%	DB	1.9	1.8	3.2	11.2	18.0	21.6	24.5	22.3	16.9	11.2	7.0	4.1	(25)	
		MCWB	1.4	1.2	1.5	6.4	11.5	14.8	17.3	16.2	13.7	10.0	6.4	3.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	3.6	6.7	11.9	18.0	19.8	21.6	20.8	17.2	13.1	9.4	7.5	(27)	
		MCDB	4.9	4.4	10.6	18.2	24.6	25.3	26.7	25.5	19.8	14.1	9.8	7.8	(28)	
	2%	WB	3.3	2.6	4.3	9.9	15.6	18.2	20.5	19.2	16.2	12.2	8.3	6.0	(29)	
		MCDB	3.6	3.2	6.8	14.8	20.0	23.2	25.6	23.2	18.5	13.0	8.7	6.5	(30)	
	5%	WB	2.3	1.9	3.0	8.4	13.7	16.8	19.3	18.2	15.3	11.2	7.3	4.8	(31)	
		MCDB	2.6	2.3	4.6	12.8	19.2	21.7	24.6	22.0	17.3	12.1	7.7	5.3	(32)	
	10%	WB	1.4	1.1	1.8	7.0	12.1	15.6	18.3	17.3	14.3	10.3	6.4	3.5	(33)	
		MCDB	1.7	1.5	2.8	10.4	17.5	20.2	23.1	20.9	16.2	11.1	6.8	4.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.7	7.2	8.6	10.0	9.1	9.1	8.9	7.7	5.6	4.2	4.6	(35)	
		MCDBR	3.9	5.0	9.1	12.9	13.5	12.1	11.7	11.6	9.2	6.2	4.3	3.9	(36)	
	5% WB	MCWBR	3.8	4.3	5.9	7.1	6.6	5.6	4.9	5.1	5.3	4.7	4.2	3.9	(37)	
		MCWBR	3.8	4.3	7.6	11.8	11.8	10.3	10.1	9.3	8.1	5.4	4.1	3.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.275	0.310	0.332	0.358	0.356	0.354	0.375	0.368	0.341	0.331	0.302	N/A	(40)	
		taud	2.136	2.195	2.255	2.357	2.423	2.473	2.434	2.452	2.518	2.398	2.214	N/A	(41)	
	Edn at Noon		473	671	788	840	871	878	846	824	788	655	437	N/A	(42)	
		Edh at Noon	42	71	93	101	102	99	101	92	73	59	40	N/A	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.25	0.81	2.30	3.84	5.21	5.41	5.23	4.04	2.46	1.08	0.30	0.14	(44)	
	RadStd		0.04	0.17	0.30	0.40	0.56	0.43	0.48	0.42	0.32	0.17	0.06	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.53	N/A	N/A	N/A	N/A	N/A	-145	-206	N/A	N/A
(47) Regional (58 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	-144	N/A	N/A	N/A

Nomenclature: See separate page